

High Level

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1. Compare type-1 and type-2 hypervisors for a cloud datacenter hosting mission-critical applications.

<u>Feature</u>	<u>Type-1 Hypervisor</u>	<u>Type-2 Hypervisor</u>
Installation	Runs directly on hardware	Runs on top of a host operating system.
Performance	Very high, low overhead	Lower due to host OS
Security	Strong isolation and smaller attack surface	more vulnerable because host OS can be attacked.
Reliability	high stable for work load enterprises	depends on host OS stability.
Scalability	supports thousands of VMs	suitable for limited VMs.
Examples	VMware ESX, microsoft hyper-v xen.	VMware workstation, Oracle virtual box

manageability:-

- Type-1: supports centralized management, clustering, live migration, high availability, and automated resource.
- Type-2: is mainly intended for desktop testing, development and training.

Justification:-

for a cloud data center hosting mission-critical applications, type-1 hypervisors are the better choice because they provide better performance, stronger security and high availability.

2.

workload isolation using virtualization architecture.

a cloud provider hosts finance, healthcare and student portal applications on the same physical server.

How virtualization provides isolation:

1. Virtual machines:

Each workload runs inside its own VM with a dedicated virtual operating system.
Failure in one VM does not affect others

2. CPU isolation:

Hypervisor allocates virtual CPUs independently.
Prevents CPU interference among workloads.

3. Memory isolation:

Each VM receives dedicated virtual memory.
Memory protection prevents unauthorized access

4. Storage isolation:

Separate virtual disks ensure independent data storage.
Encryption can further secure sensitive data.

5. Network isolation:

Virtual switches, VLANs and SDN separate network traffic.

Finance traffic cannot directly communicate with healthcare systems.

Conclusion:-

Virtualization ensures secure workload isolation through compute, memory, storage and network separation.

3. VM host utilization analysis.

Given Snapshot

CPU utilization = 92%

memory utilization = 88%

Storage I/O Latency = 35ms

Average VM Boot = 140s

Bottle neck analysis:

CPU (92%)

CPU is nearly saturated

Slower VM execution

Reduced application responsiveness.

Memory (88%)

• memory usage is very high.

• possible swapping or ballooning may occur.

VM Boot Time (140s)

• Normal enterprise VM boot time is typically 30-90sec

• Long boot time suggests storage and CPU bottlenecks

Overall Recommendation:-

Problem

likely cause

corrective action

CPU 92%

CPU contention.

Add CPU's

memory 88%

High memory pressure

Increase RAM

Storage latency

Disk bottleneck

SSD / Nvme

Boot time

Storage + CPU

Improve storage.

4. Software defined datacenter (SDDC) vs traditional datacenter.

Traditional Datacenter:

- Hardware resources are manually configured.
- separate management for compute, storage and networking.
- Scaling requires physical hardware installation.
- Limited automation.

Software defined datacenter.

All infrastructure resources are virtualized and managed through software.

* Computer virtualization:

- creates multiple vms on shared hardware.
- supports live migration.
- enables automatic resource allocation.
- improves server utilization

* Storage virtualization

- combines multiple storage devices into a logical storage pool.
- enables: thin provisioning, snapshots, replication.

* Network virtualization

- uses software - defined networking (SDN)
- Network policies are software controlled.
- supports: virtual switches, dynamic routing.

Conclusion.

SDC improves agility through:

- Automated infrastructure provisioning
- Better resource utilization.

4.

5. Identify mismatches in a multi cloud federation.

Problem Analysis:

Different cloud providers use different:

- user identity formats
- Authentication mechanisms
- access control policies.
- Attribute naming conventions.

* Secure Identity Federation design.

1. federated identity:

- maintain one trusted identity across all cloud providers
- user authenticate once and access multiple clouds.

2. Identity provider (IdP)

- central Authority authenticates users
- cloud providers acts as service providers

3. privacy protection:

- shares only the minimum required user attributes.
- encrypt identity assertions.
- use consent based attribute release.
- Apply role based or attribute based access control.

Conclusion:-

a federal identity architecture using trusted identity provider, standardised protocols secure attribute mapping, privacy-preserving access control resolves identity mismatches while ensuring and seamless authentication across multiple cloud providers.